DSA06 Data Handling and Visualization | Tableau Introduction Activity (25 marks) | Slot D

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Your data file: D_2.csv | Tool: Tableau Public (free, public.tableau.com) | Total: 25 marks | Time: two sessions

The file holds 12 months of 2025 for four regions (Telangana, Tamil Nadu, Gujarat, Andhra Pradesh). Fields: Month, Region, Sales (Rs lakh), Profit (Rs lakh), Customers, Discount_pct. First rows of your file:

Month	Region	Sales	Profit	Customers	Discount_pct
2025-01-01	Andhra Pradesh	109.1	12.1	1085	8.9
2025-01-01	Gujarat	79	9.2	1072	4.9
2025-01-01	Tamil Nadu	45.7	3.8	425	6.4
2025-01-01	Telangana	57	7.7	576	6.8
2025-02-01	Andhra Pradesh	111.4	10.7	1080	13
2025-02-01	Gujarat	73.3	0.8	990	19.1

T1. Connect and check [3]

Open Tableau Public, connect to your CSV file (Text file), and set Month as a Date. Report the number of rows, which fields are dimensions and which are measures, and the total Sales and total Profit.

T2. Bar chart [4]

Build a bar chart of Sales by Region, sorted from highest to lowest, with labels on the bars. Name the top and the lowest region and give the top region's share of the total sales (%).

T3. Line chart and trend [5]

Build a line chart of total Sales by Month. Add a linear trend line (Analytics pane) and a 3-month moving average (right-click SUM(Sales) > Quick Table Calculation > Moving Average). Report the peak month and its value, the moving average at June, and the trend slope per month (from Describe Trend Model).

T4. Scatterplot [4]

Build a scatterplot of Discount_pct (columns) against Profit (rows), one point per row (turn off aggregation), coloured by Region, with one overall trend line. Report the slope, r (the square root of R-squared with the sign of the slope) and say in one sentence what it means for the business.

T5. Filled map [3]

Build a filled map of Profit by Region (double-click Region; Tableau should recognise Indian states). Name the region with the lowest total Profit and compute its profit margin (Profit / Sales, %).

T6. Dashboard with a filter [4]

Place the bar chart, the line chart and the map on one dashboard and make the bar chart a filter (Use as Filter). Click the region with the SECOND-highest sales and report that region's total Sales, total Profit and profit margin (%). Publish to Tableau Public and paste the link.

T7. Compare with R [2]

Write two real differences between doing the T3 line chart in Tableau and doing it in R (ggplot2).

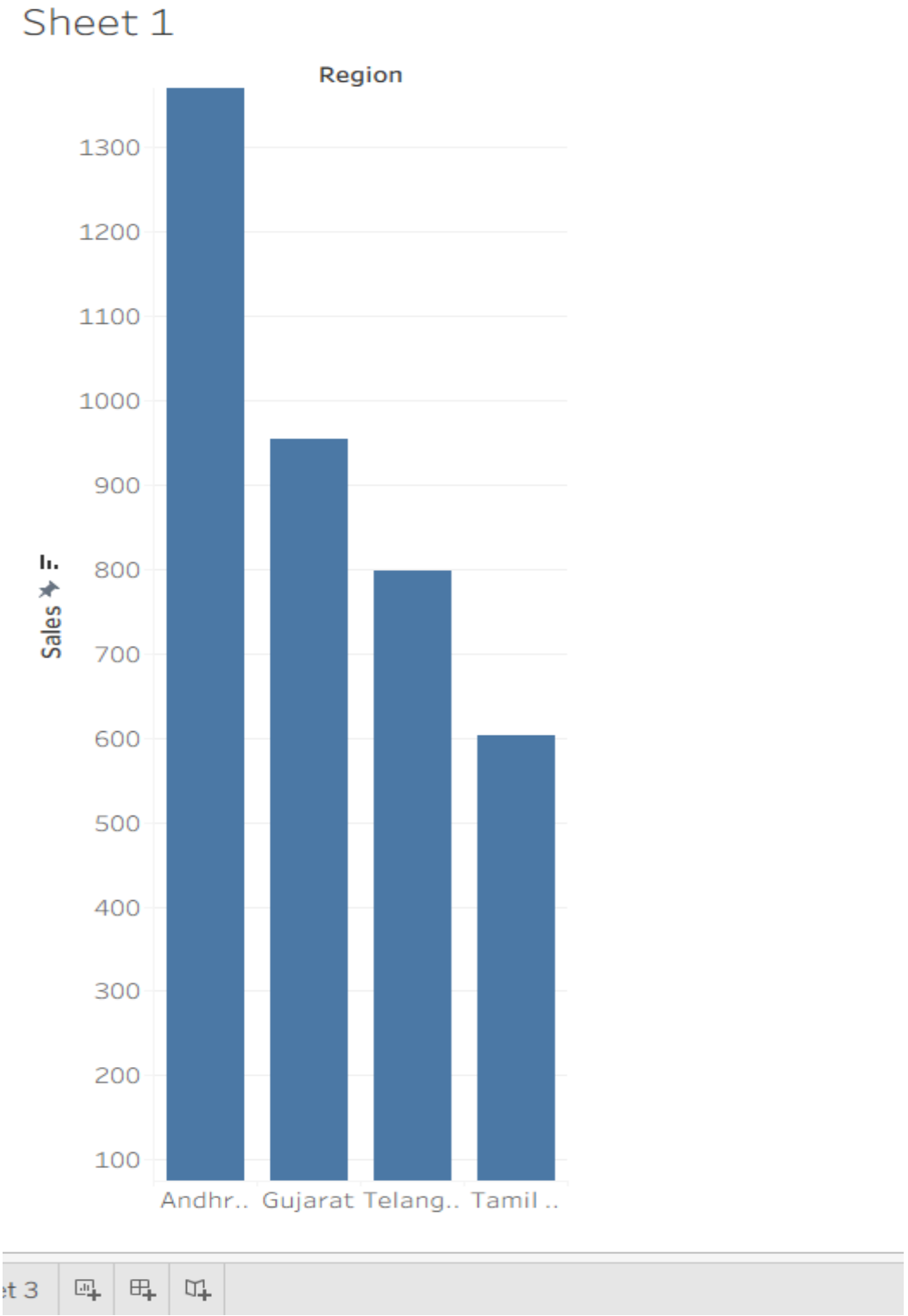
Marking: correct values from YOUR OWN file, readable charts with titles and labelled axes, and clear sentences. Everyone answers the same seven tasks; only the data in your CSV file are different.

SOLUTION

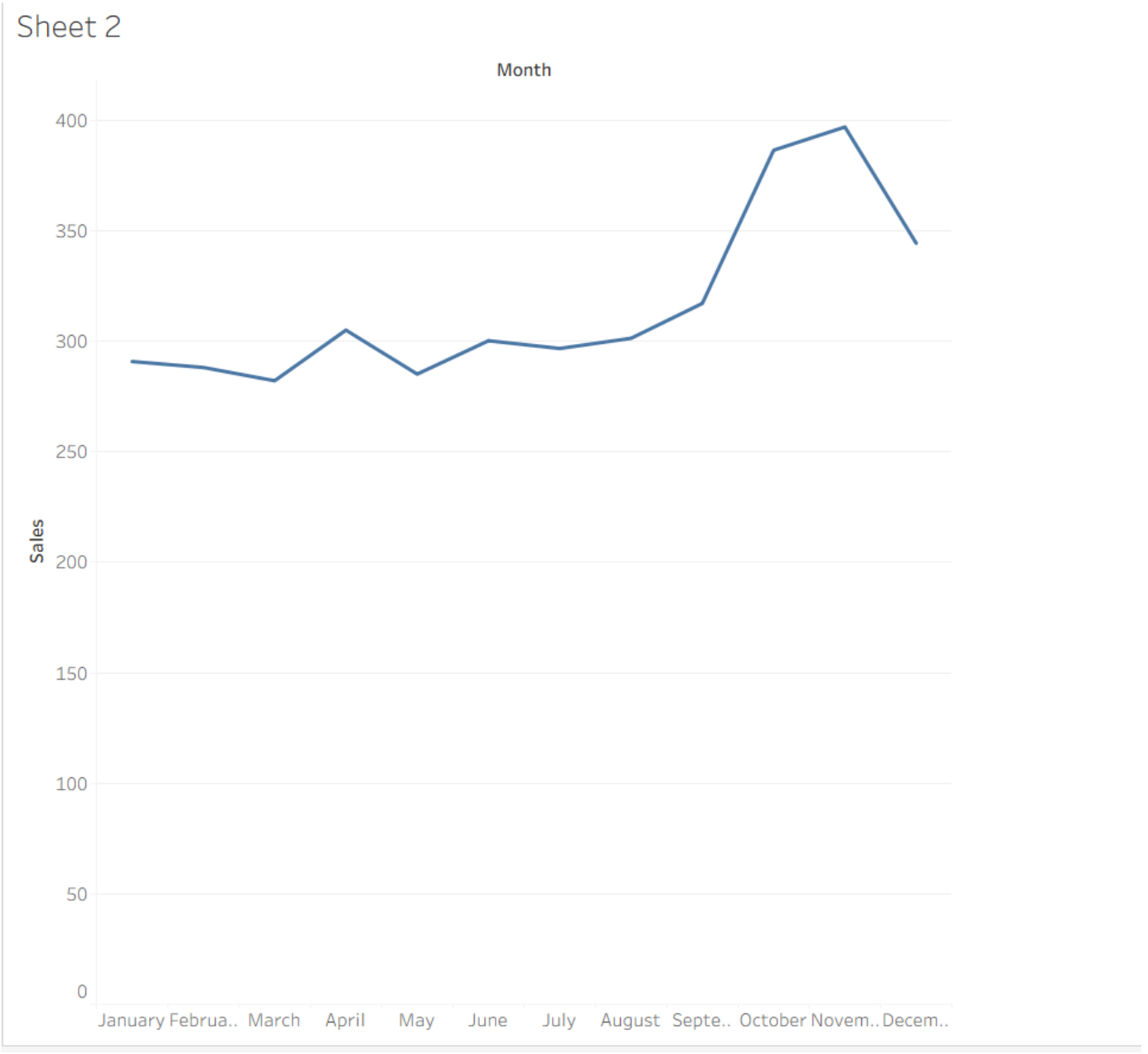
T1. Connect and check — 3 marks

1. *Open Tableau Public and select Connect → Text File.*
2. *Open D_2.csv.*
3. *Check that Month is recognised as a Date.*
4. *Go to a worksheet and drag the required fields to the view.*

T2. Bar chart — 4 marks



T3. Line chart and trend — 5 marks



T4. Scatterplot — 4 marks

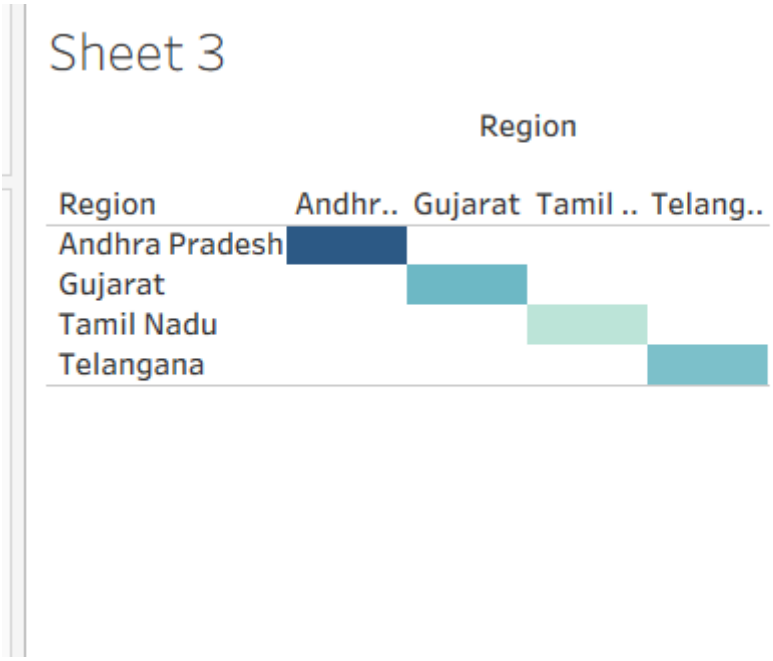
1. *Drag Discount_pct to Columns and Profit to Rows.*
2. *Select Circle in Marks.*
3. *Drag Region to Color.*
4. *Turn off Analysis → Aggregate Measures to show one mark per row.*
5. *Open Analytics and add one linear trend line.*

Report the slope and correlation coefficient r.

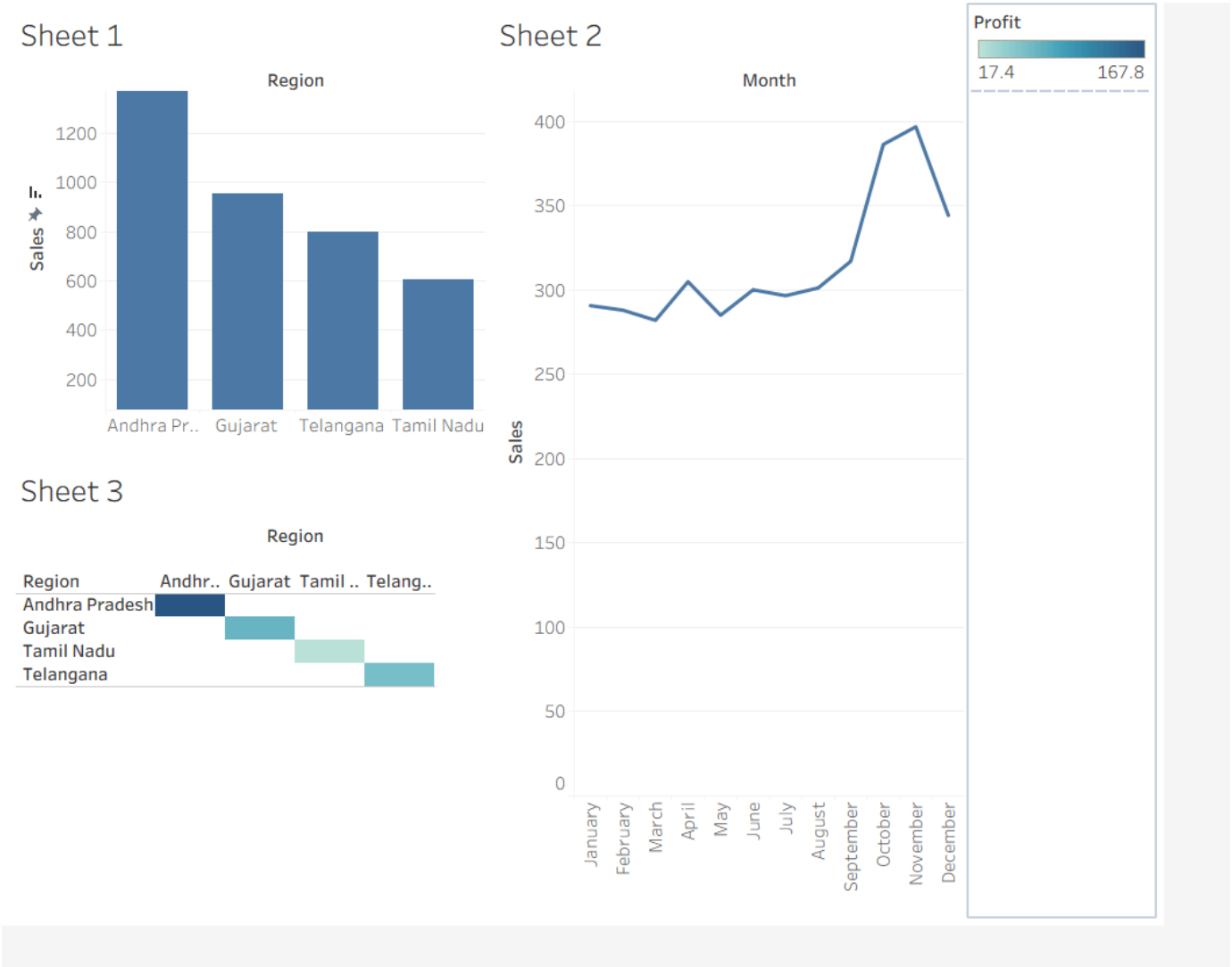
\[r=\operatorname{sign}(\text{slope})\sqrt{R^2} \]

Business interpretation: A negative slope means higher discounts are associated with lower profit; a positive slope means higher discounts are associated with higher profit. This indicates an association, not necessarily causation.

T5. Filled map — 3 marks



T6. Dashboard with a filter — 4 marks



T7. Compare Tableau with R — 2 marks

Tableau	***R (ggplot2)***
The line chart is created interactively by dragging fields onto Rows and Columns.	*The line chart is created using code, such as ggplot() and geom_line().*
Moving averages and trend lines can be added through Tableau's built-in table calculations and Analytics pane.	*Moving averages and trend lines are calculated or added using R code and appropriate functions or packages.*